Chp 1

- Morals : religious views on right / wrong.
- Ethics: rules / codes of conduct that define acceptible behaviour in society
- Law: determines what we are permitted or not permitted to do.
- Computer ethics : guidelines for morally acceptable use computers.

Morals, Ethics & Law

Aspect	Morals	Ethics	Laws
Based on	Religion / personal belief	Society / profession's code of conduct	Government rules
Nature	Individual & internal	Collective & professional	Enforceable by police / courts
Example	Believing abortion is wrong	IT professional code against piracy	Abortion legalized in some countries

Contrast: A lawyer may ethically defend a guilty client (professional duty), though it may seem immoral to some.

Why — Ethics in IT

- affects governance, work & personal life
  - ↳ ethical decisions matter
  - ↳ involve balancing company interests & individual rights
- Issues include:
  - ↳ Privacy (employee monitoring, email usage)
  - ↳ Data security (protection against manipulation / leaks)
  - ↳ Global concerns (cross-border data, cybercrime)

Example

- A company tracks employee browsing history to prevent misuse, but it raises ethical concerns about piracy.

-Company installs software that tracks employee keystrokes.

Ethics for IT Professionals

Characteristics of professionals:

↳ Advanced knowledge & creativity

↳ Independent judgement

↳ Commitment to lifelong learning

↳ Contribution to society

Professional Code of Ethics

↳ Sets out goals & rules for members.

↳ Guides behaviour beyond only following the rules.

Common Ethical Issues for IT Users

Software Piracy: illegal copying of software

Inappropriate Use of IT: browsing, gaming / sharing offensive content at work, wasting company resources.

Inappropriate Sharing of Information: leaking confidential business / customer data.

Certifications in IT

Proof of skills, often boosts career & salary.

Vendor Certifications: Specific to a company's products

↳ Cisco, Microsoft, AWS

Industry Association Certifications: Broader, requires exams & ongoing learning

↳ $(ISC)^2$, ISACA

Example

- AWS Cloud Certification

↳ shows ability to manage cloud services.

- A Cisco-certified network engineer might be hired faster bcs certification reassures employers of basic competence.

Chp 2

Cybercrime : Illegal activity using the Internet

- A computer can be:
  - ↳ Weapon (used to launch attacks)
  - ↳ Target (hacked)
  - ↳ or Both

Types	Description
Malicious harm	Stealing data from company.
Accidental harm	An employee accidentally spreading a virus.

Computer User Expectations

- Technology allows faster, cheaper data storage.
- More data collected, easier to track users
- Adult trails = record every modification

Example:

- Sharing your office Wi-Fi password with a friend can risk company data leaks
- Using fingerprint login is convenient, but if stolen cannot be "reset" like a password

Problems

- Users share login credentials → unauthorized access
- Increasing complexity → more vulnerabilities
- Rapid tech changes → harder to keep secure

Reliance on Commercial Software

- Weakness in software = exploits
- Companies release patches to fix flaws, but if users don't install = risk of breach
- Most cyberattacks happen bcs people don't install patches in time.

Types of Security Attacks

Types	Examples	Examples
Passive Attacks (stealing info, no dmg) (hard to detect) (silent spying)	• eavesdropping ↳ (listening to calls/emails) • traffic monitoring ↳ (observing communication patterns)	• Hacker listens to your Wi-Fi traffic.
Active Attacks (tampering/disrupting) (visible damage)	• Masquerade ↳ pretending to be someone else. • Replay ↳ resending captured messages • Message alteration ↳ changing/delaying messages • Denial of Service (DoS/DDoS) ↳ overwhelming a system w fake requests	• Hacker floods your Wi-Fi router so it crashes.

Common Malware & Threats

Types	Desc	Examples
Virus	attaches to files, spread when opened	• Phishing scam ↳ A fake "bank email" asking you to click a link. • Clicking a fake "free game" download (Trojan) installs a botnet client & your PC joins a DDoS attack.
Worm	spreads automatically.	
Trojan Horse	disguised as legit software, cannot self-replicate	
Botnet	network of infected devices controlled by hacker	
Rootkit	hides attacker's presence	
Spam	unwanted messages	
Phishing	tricks users into revealing sensitive info	

Hackers & Crackers

Types	Desc
Ethical hackers (white hats)	Tests systems legally, help improve security (eg: CEH certified)
Crackers (black hats)	Malicious hackers who break systems for personal gain.

Malicious Insiders

- employees / ex-staff abusing access.
  ↳ often are colluding w outsiders for fraud.

Cybercriminal Categories

Type	Desc	Examples
Cybercriminals	- financially motivated ↳ (stealing credit card data, funds)	• Stealing credit cards.
Hacktivists	politically / socially motivated	• Defacing a government website to protest a policy.
Cyber-terrorists	- aim to cause large-scale harm ↳ (eg: hacking air traffic systems)	• Attacking air traffic systems to cause chaos.

Chp 6

- General Security Risk Assesment
  - ↳ A systematic way to identify risks to an organization's IT systems (hardware, software, databases, network)
  - ↳ Helps decide amout of money & time should be invested in protecting the organization.
- Key concepts
  - Assets
    - ↳ anything valuable (eg: servers, data, applications)
  - Failure case
    - ↳ any harmful event (eg: virus infection, DDoS attack)
  - Risk Formula
    - ↳ Risk level = Probability of Threat x Impact of Threat

Trustworthy Computing

↳ An approach ensuring technology is secure, private, reliable & ethical.

↳ To build confidence in computing for customers & business.

Core Priciples	Desc	Example
Security	System resists attacks.	Firewalls & antivirus software
Privacy	User control their own data	GDPR compliance in Europe.
Reliability	Systems work consistently	Online banking systems available 24/7.
Business Integrity	Companies acting ethically	Transport billing by cloud providers.

CIA Triad

Triad	Confidentiality	Integrity	Availability
Desc	• Only authorized ppl can access data.	• Data can't be improperly altered.	• Systems must be accesible when needed.
Example	- Password protected medical records.	- Digital signature to prevent tampering.	- E-commerce sites working during Black Friday.

IT Security Policy

↳ A written plan on how a company protects its IT assets.

↳ Exp:

- password policies (eg: must change every 90 days)
- Remote access rules (VPN required)
- Data backup & disaster recovery procedures

International Standards (ISO 27001) (ISMS)

↳ A global standard for Information Security Management System

↳ Exp of covered areas:

- Access control (who can access what)
- Cryptography (data encryption)
- Human resource security (training employees)
- Compliance w laws

↳ A company certified under ISO 27001 can prove to customers that their data is handled securely.

NIST Cybersecurity Framework (CSF)

↳ created by: U.S. National Institute of Standards & Technology

↳ to help organizations identify, protect, detect, respond & recover from cyber threats.

Functions	Desc
Identify	• know your assets & risks
Protect	• safeguards (firewalls, training)
Detect	• find security incidents early
Respond	• take action to contain impact
Recover	• restore systems & operations

Organizational Profiles:

↳ Current Profile: Where we are now

↳ Target Profile: Where we want to be

CSF Tiers

↳ Levels of implementation (Tier 1 = Low, Tier 4 = adaptive & advanced)

↳ Exp: A bank might be at Tier 3 (respectable, well-documented security) & aim for Tier 4.

Chp 7

Software Development Methodology

↳ A structured way to create software efficiently & with quality.

↳ Popular Methods:

Waterfall: Step-by-step, no going back.
Good for projects with fixed requirements.
Exp: Government tax software system

Agile: Flexible, iterative, feedback driven.
Good for changing requirements
Exp: Mobile app that updates every 2 weeks.

Cost of fixing errors

↳ Fixing a bug early (design stage) is much cheaper than fixing it after release.

Quality Assurance (QA) & Standards

↳ QA = Ensures software is "fit to use"

↳ PDCA Cycle (Deming Cycle):

Cycle	Desc
Plan	Define objectives
Do	Implement
Check	Monitor results
Act	Improve process

Exp: If user complaints arise, QA reviews the process & updates testing.

Software Testing

↳ Ensures the software works as expected.

Types	Desc
Manual Testing	• Human testers simulate real usage.
Automation Testing	• Tools / scripts automatically test functionality.
Dynamic Testing	• Runs the program to check behaviour. ↳ Black-box: Tester doesn't know the code, only checks input & outputs. ↳ White-box: Tester knows code structure & checks all logic paths.

Exp: • Black-box → Testing login form with valid / invalid inputs.
• White-box → Checking every line of a function runs at least once.

Software Quality & Assurance Techniques

Attributes	Desc
Scalability	• Runs on many devices.
Usability	• Easy for all users.
Reusability	• Code components reused in other projects.
Correctness	• Meets specifications
Maintainability	• Easy to fix & update.

Exp: A mobile payment app should be usable by all (simple UI), scalable (works on Android & iOS), & maintainable (bugs fixed quickly).

Chp 8

Introduction to IP

↳ Creations of the mind (inventions, art, designs, brand names)

↳ Can be more valuable than physical assets.

↳ Protects innovation & creates competitive advantage.

Intellectual Property Rights

↳ Rights given to creators to protect & benefit from their work.

↳ Prevents unauthorized use.

↳ Exp: patents, trademarks, copyrights, trade secrets.

Types of Intellectual Property

a) Patents

↳ Protect inventions

↳ Governed by Patent Act 1983.

↳ Requirement: New, inventive & industrially applicable

↳ Valid for 15 years from grant

↳ Exp: Apple patents iPhone touchscreen technology

b) Trademarks

↳ To distinguish good / services from competitors.

↳ Governed by Trademark Act 2019.

↳ Types of Trademarks (distinctiveness spectrum)

- Generic (weakest: "Milk")
- Descriptive ("Fast Cleaners")
- Suggestive ("Netflix")
- Arbitrary ("Apple" for computers)
- Fanciful / Coined ("Kodak")

* The Nike ✓ logo

c) Copyrights

↳ Protects artistic / literary works.

↳ Copyright Act 1987, updated in 2012 & 2022.

↳ Lasts life of creator + 50 years.

↳ Based on international conventions (Berne Convention)

* Exp: song, novel / software code

d) Trade Secrets

↳ Confidential info giving the business an advantage (eg: Coca-cola receipe)

↳ No specific law, but protected under contracts and common law

↳ Protection via NDAs, access control, employee agreements

Issues in IP

Issues	Desc
Plagiarism	· Copying without credit. ↳ Common in schools, writing & software.
Reverse Engineering	· Disassembling products to copy designs. ↳ Legal in some context but ethically debated.
Open Source & Closed Source	· Open: free to use & modify (Linux) · Closed: propriety (Windows)
Competitive Intelligence (CI)	· Gathering patent / market data to monitor competitors.
Trademark Infringment	· Using another's mark illegally (fake Adidas shoes)
Cybersquatting	· Registering domain names of famous brands for profit or fraud.

General Security Risk Assessment

Steps	Desc
Identify Assets	· Prioritize worrying abt IT properties that promote the org's agenda & objectives.
Specify Loss Events	· Identify potential loss cases, risks & threats (DDOS atk/ insider theft)
Frequency of Events	· Estimate num of incidents & probability of each threat. (insider theft may occur more often)
Impact of Events	· Determine severity of each hazard. Risk Tolerance = Probability x Impact
Options to Mitigate	· Decide if each hazard can be mitigated so that it's less likely / has smaller effect.
Feasibility of Options	· Evaluate viability of proceeding w the mitigation options
Cost/Benefit Analysis	· Conduct cost-benefit study & guarantee it's profitable
Decision	· Determine whether to implement a particular countermeasure.